

Vidath Dassanayake

COMPUTER SCIENCE UNDERGRADUATE | SOFTWARE ENGINEERING

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building and deploying a full-stack TypeScript SaaS platform using Next.js, Node.js/Express, and PostgreSQL on GCP as part of a 6-person Agile team. Integrated a fine-tuned LLM into the platform and have experience with unit testing (JUnit), API testing, CI quality gates, and AI-assisted development tools.

EDUCATION

Informatics Institute of Technology (IIT) – BSc (Hons) Computer Science · Year 3 of 4 · Expected graduation: 2024 – Present 2028

Modules – Object-Oriented Programming (Java/Python, JUnit unit testing), Databases, Client-Server Architecture, Web Development

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Java, Python, C# (Unity), HTML, CSS, OOP
Frameworks & Libraries:	React, Next.js, Node.js, Express.js, Prisma ORM, PyTorch
Databases:	PostgreSQL, Redis, SQL
Testing:	Unit testing (JUnit, Python), Postman (API testing), manual functional and cross-browser testing, CI quality gates (ESLint, type-checking, Docker build verification)
Cloud & DevOps:	Docker, Kubernetes, GKE, GCP, Terraform, Helm, GitHub Actions, Kong
AI-Assisted Development:	Gemini Pro, Claude Code, GitHub Copilot
Other:	REST APIs, Agile/Scrum, Git, Linux, Bash, Unity 6, Blender

PROJECTS

OreanStudio (Orean 360 Platform)

2025 – 2026

Technologies: Next.js, React, Node.js 20, Express.js, TypeScript, Zod, PostgreSQL, Prisma ORM, BullMQ, Redis, FFmpeg, Kubernetes, GKE, Terraform, Helm, Kong API Gateway, GitHub Actions

- Built a Next.js 16 frontend and a Node.js 20 / Express 5 REST API in TypeScript (Prisma ORM, JWT auth, Zod validation) on a 14-model PostgreSQL schema.
- Built the post-creation, Kanban board, due-date update and calendar date-range endpoints, plus social-token disconnection.
- Implemented six OAuth integrations (Google, YouTube, Meta, LinkedIn, TikTok, WhatsApp), with BullMQ workers for scheduled publishing (retries with exponential backoff, publish audit logs) and FFmpeg video transcoding into five platform presets.
- Automated Prisma migrations and Helm deployments to GKE through GitHub Actions CI/CD (with atomic rollback), and implemented KEDA-driven autoscaling for BullMQ workers.
- Diagnosed and resolved 23 documented issues across two projects (symptom, root cause, fix), including unprotected dashboard routes and an exposed API key purged from git history with git filter-repo.
- Collaborated with a 6-member engineering team using Agile/Scrum practices to design and deliver the platform.

GEO Engine (Zero-Cost LLM Pipeline)

2025 – 2026

Technologies: Python, PyTorch, Unsloth, Qwen2.5, LoRA, vLLM, Firecrawl, Playwright, Docker Compose, WSL2, Modal

- Built an end-to-end LLM pipeline (data collection, fine-tuning, serving) for Generative Engine Optimisation, which optimises content for LLMs like ChatGPT and Gemini.
- Collected 377 programming posts from 15 subreddits (score ≥ 100 , upvote ratio $\geq 85\%$) via the Arctic Shift archive using a rate-limited Python pipeline and self-hosted Firecrawl in Docker, then cleaned them to 374 records (minimum-length filter, deduplication, spam and all-caps filtering, PII stripping), formatted as ChatML and split 299/37/38 into train, validation and test sets.
- Fine-tuned Qwen2.5-7B-Instruct with LoRA (Unsloth, 4-bit) on a free Google Colab T4 GPU; training loss fell from 1.84 to 0.96 over 3 epochs at \$0 compute cost.
- Served the merged model with vLLM on Modal (A10G GPU, scale-to-zero after 120 seconds idle) behind an OpenAI-compatible API used by the Orean backend, within Modal's free credit tier.

Modly Blender Extension

2025

Technologies: Python, Blender Python API (bpy), FastAPI

- Developed a Blender 5.2 add-on integrating Modly's local AI 3D generation via a custom Node Graph, with a dynamic node tree generated from JSON manifests and an isolated FastAPI backend to prevent GPU conflicts.

CERTIFICATIONS

IBM SkillsBuild: [Cloud Computing Fundamentals](#)

Cisco: [Python Essentials 2](#)

Boot.dev: [Docker](#), [Git](#), [Linux](#), [SQL](#), [Python](#), [OOP in Python](#)

DataCamp: Java, Java OOP, Git, GitHub

LANGUAGES

English:	Fluent
Sinhala:	Native